

Translations



1 Consider the point $(8, 6)$.

- a Plot the point on the coordinate plane.
- b Determine the new coordinates if the point is translated 2 units down.
- c Determine the new coordinates if the point from part (b) is translated 6 units to the left.

2 Consider the point $(4, 5)$.

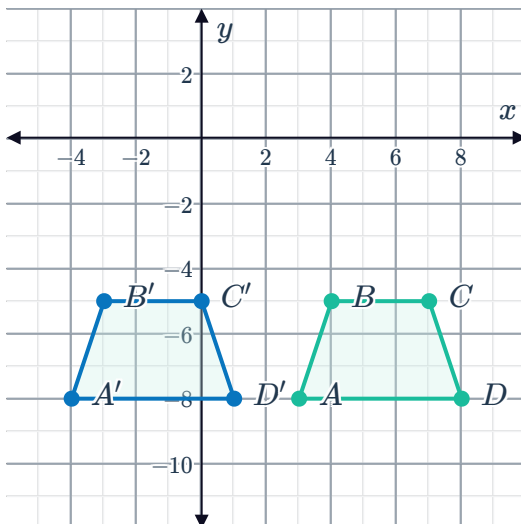
- a Plot the point on the coordinate plane.
- b Determine the new coordinates if the point is translated 2 units up.
- c Determine the new coordinates if the point from part (b) is translated 2 units to the right.

3 Plot the following translations:

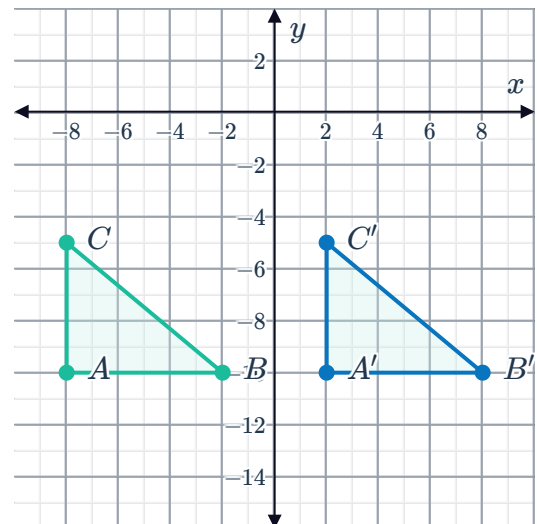
- a The point $(-8, 9)$ is translated 18 units to the right.
- b The point $(-6, 5)$ is translated 12 units to the right and 7 units down.
- c The point $(6, -2)$ is translated 15 units to the left and 6 units up.
- d The point $(8, 5)$ is translated 11 units to the left and 9 units down.

4 Describe the transformation that occurs in each graph:

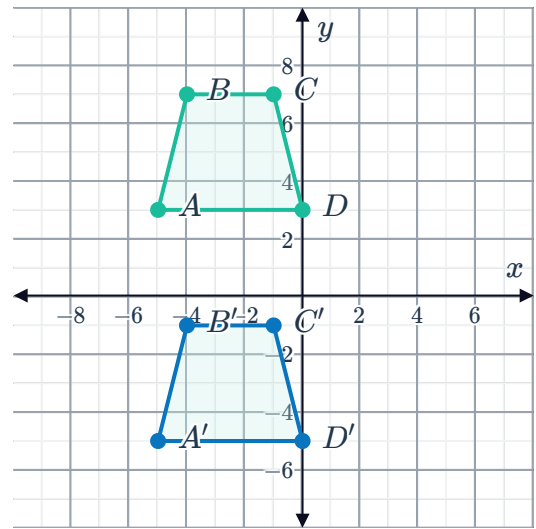
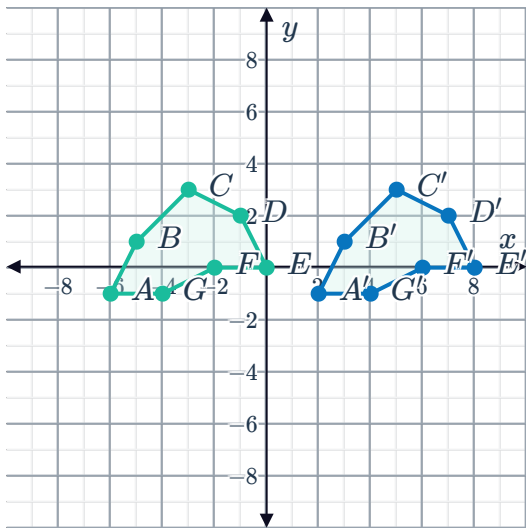
a



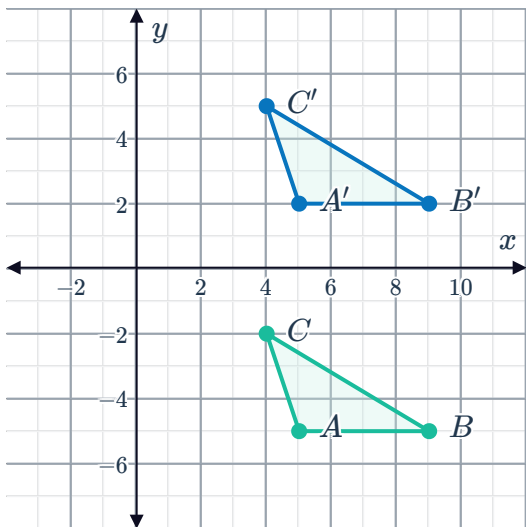
b



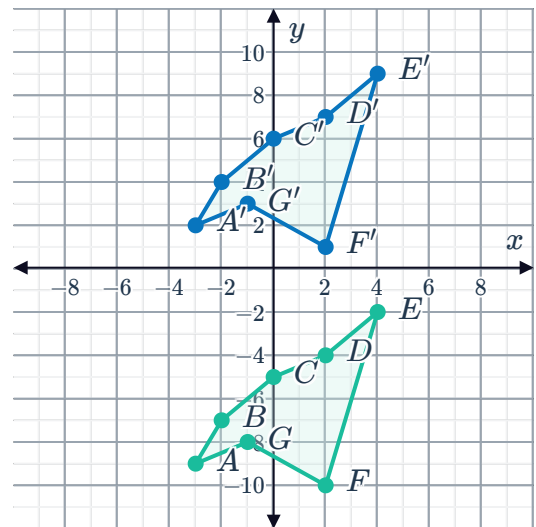
c



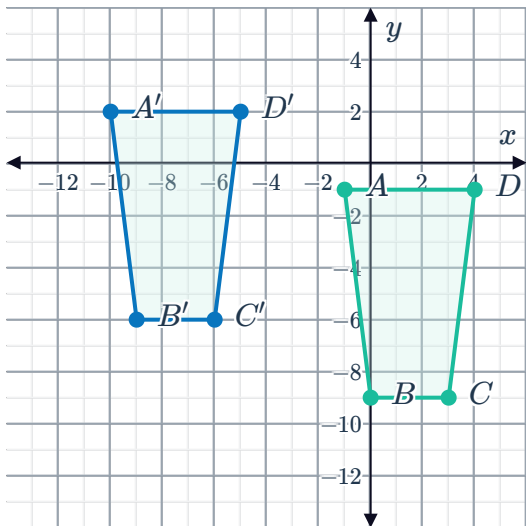
e



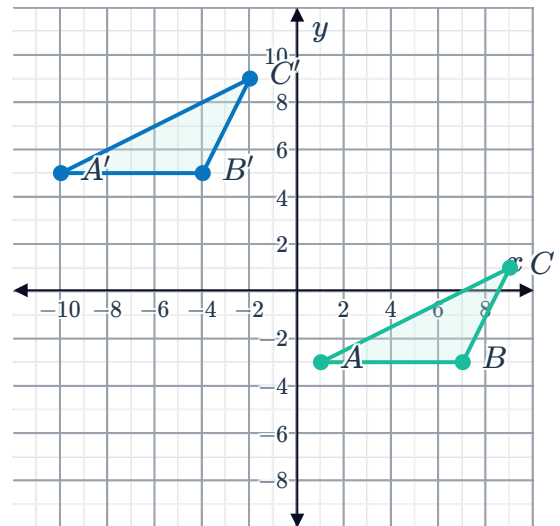
f



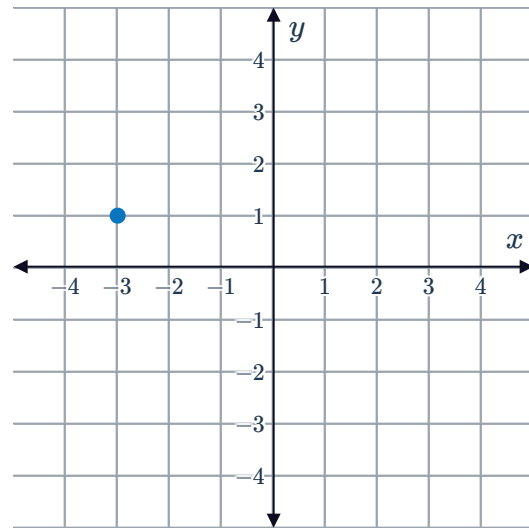
g



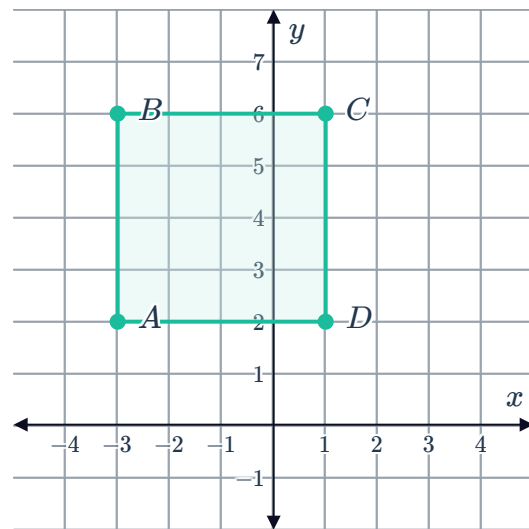
h



- 5 The point indicated on the coordinate plane is translated 4 units up. Determine the coordinates of its new position.



- 6 Quadrilateral $ABCD$ is to be translated 2 units up and 4 units to the left. Determine the coordinates of the vertex A' of the translated rectangle.

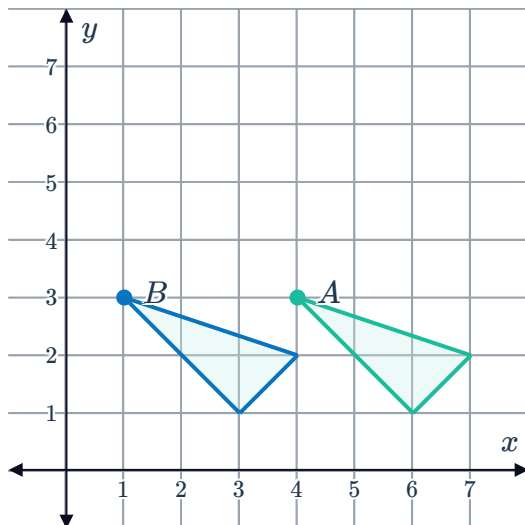


- 7 Determine the coordinates of each of the following translations of the point $A(3, -4)$:
- a Point A is translated 5 units to the right.
 - b Point A is translated 5 units to the left.
 - c Point A is translated 5 units upward.
 - d Point A is translated 5 units downward.

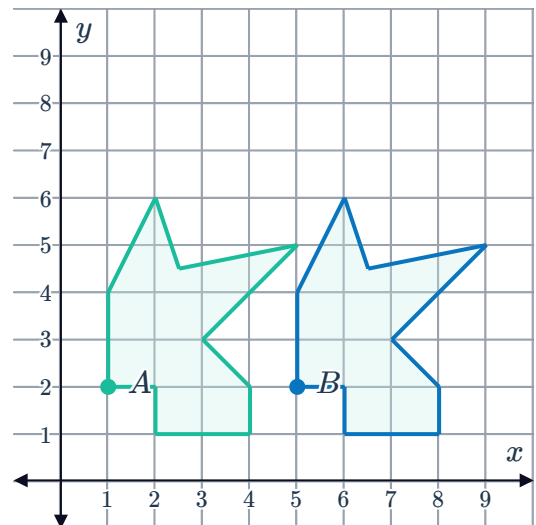
8 Determine the translation from polygon B to polygon A .



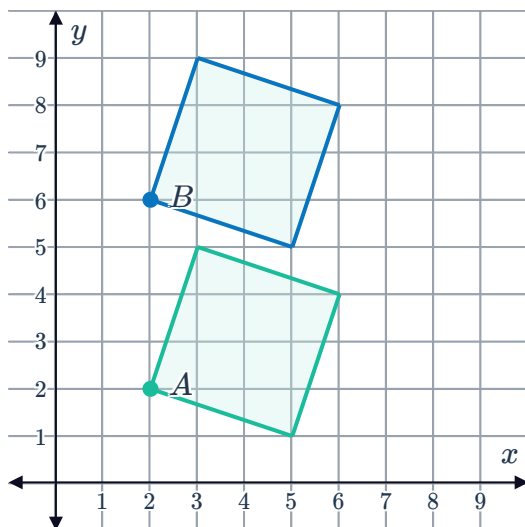
a



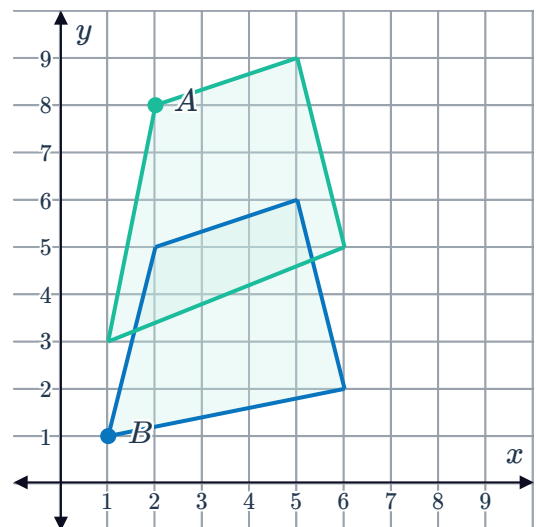
b



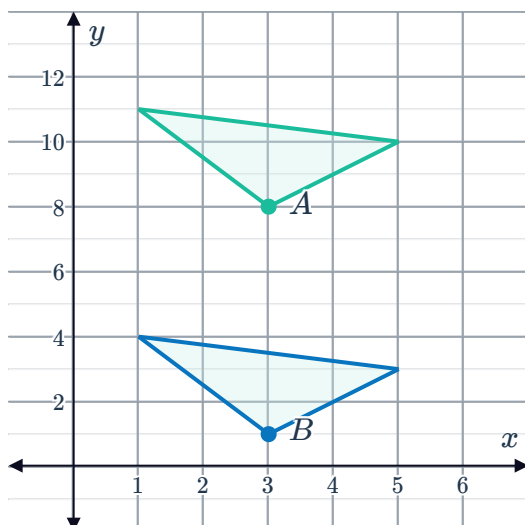
c



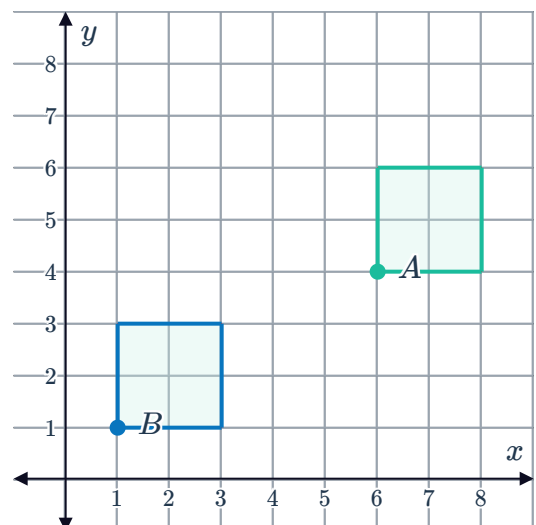
d



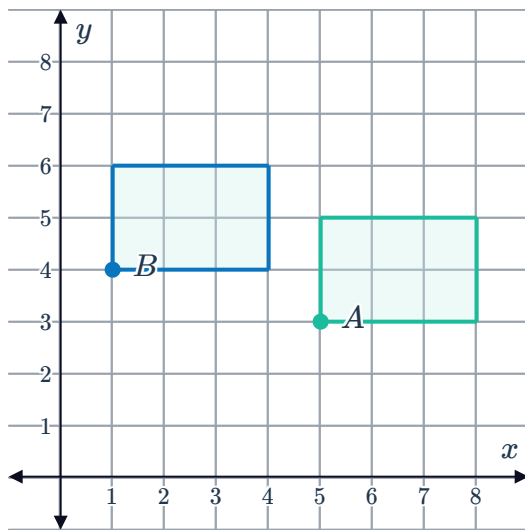
e



f

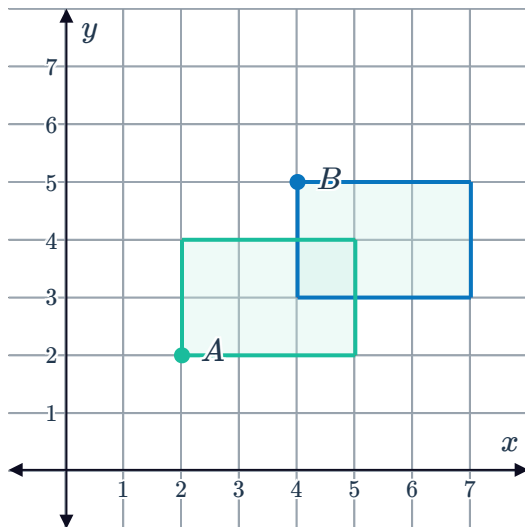


g

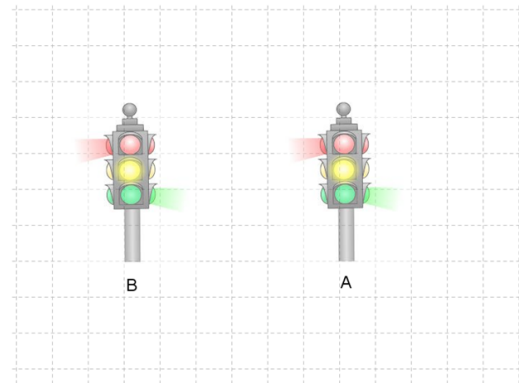


9 Determine the translation from figure *A* to figure *B*.

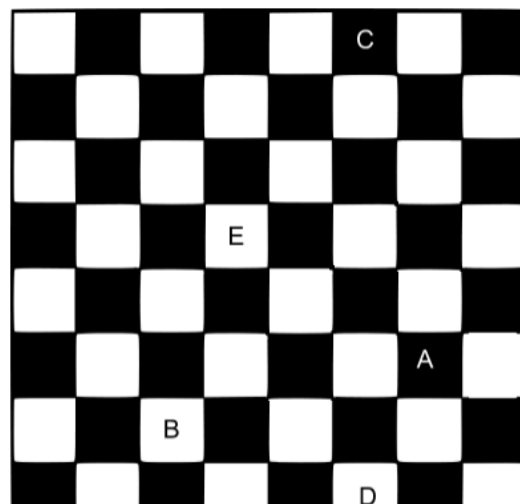
a



b



10 On a chessboard, a knight must move in an *L* shape. That is, 1 unit in any direction and then 2 units in a perpendicular direction. If a knight is at square *E* now, what labelled square could it have been two moves ago?

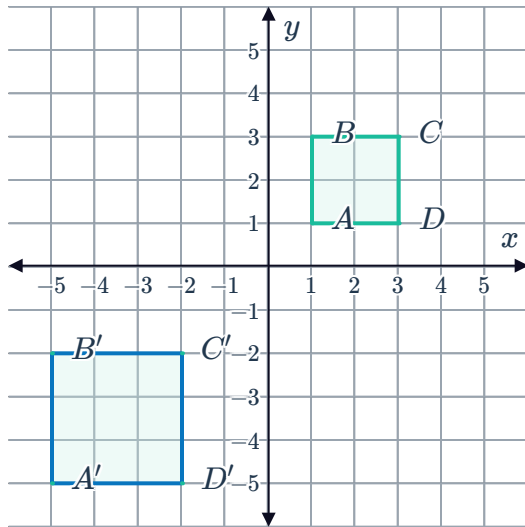




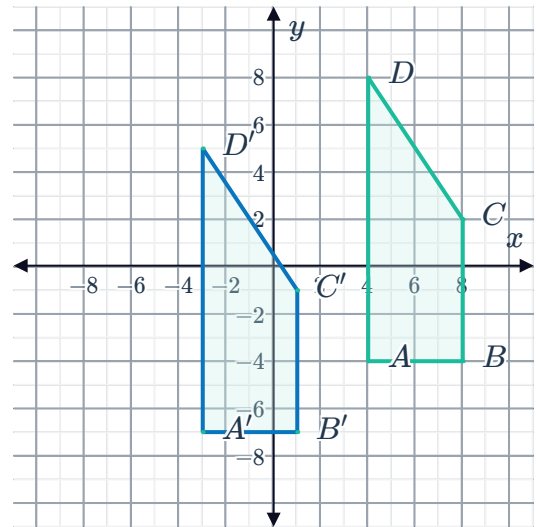
Additional questions

- 11** Describe the image produced as a result of translating a figure 4 units to the left.
- 12** Identify whether each graph shows a translation from the preimage to the image. Explain your reasoning.

a



b

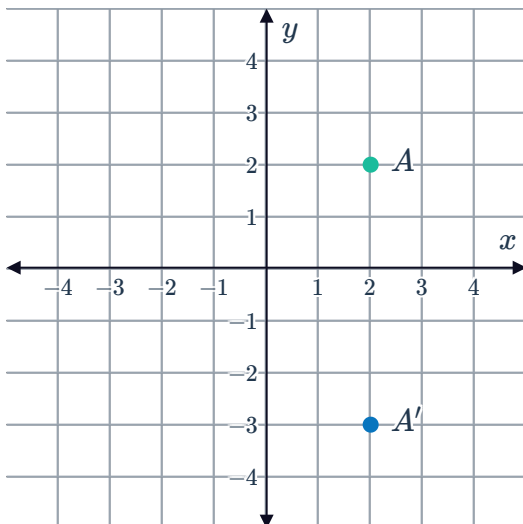


- 13** Consider the point $(8, 6)$.
- a** Plot the point on the coordinate plane. Label the point D .
- b** Translate the point 6 units to the left and 2 units down. Plot the point and label it D' .

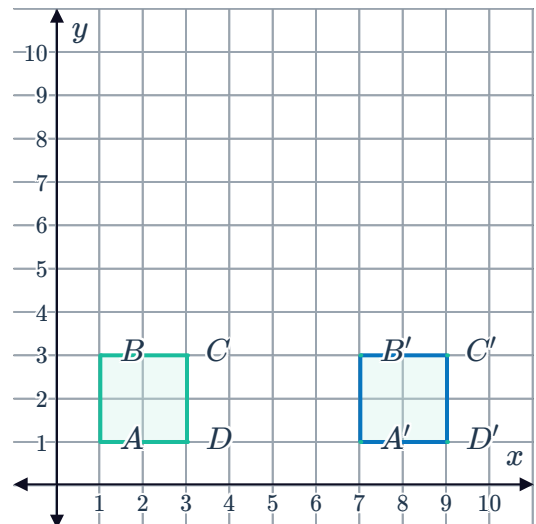
14 Describe the transformation that occurs in each graph:



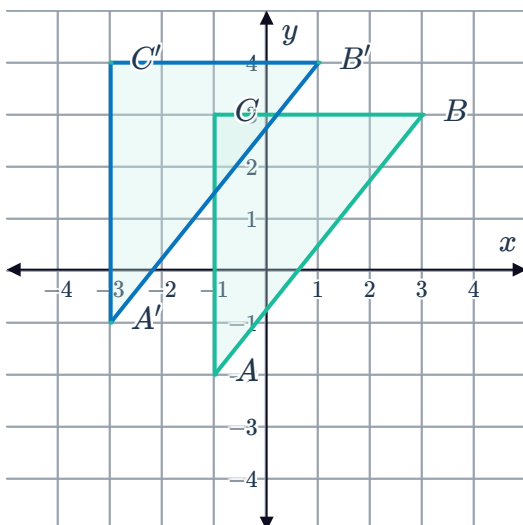
a



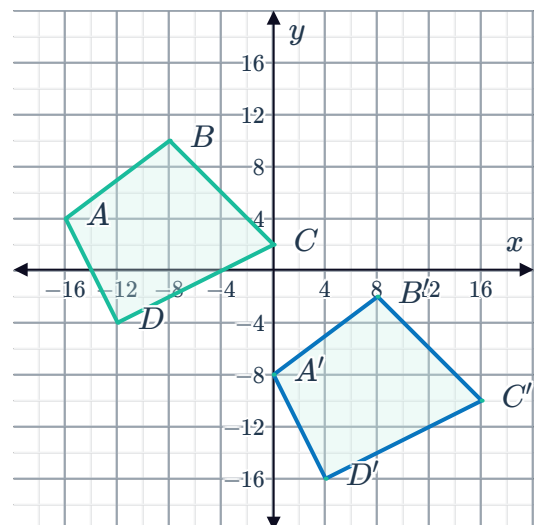
b



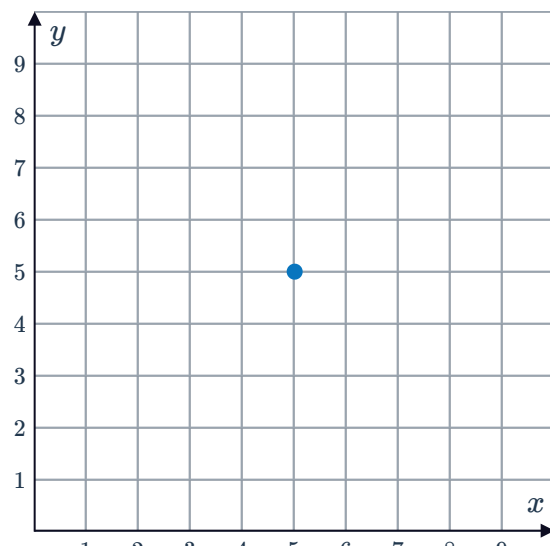
c



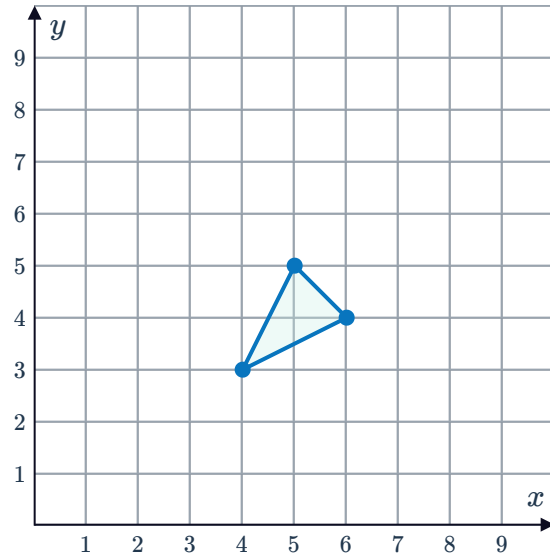
d



15 The point shown is the image produced by the translation: $(x, y) \rightarrow (x - 2, y)$. Plot the starting position of this point before it was translated.



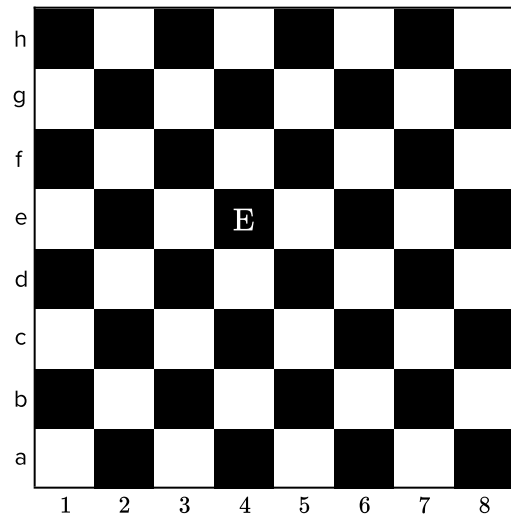
- 16 The image shown is produced by translating the triangle 3 units to the right and 4 units down.
Plot the starting position of the triangle before it was translated.



- 17 Point A is translated 3 units down and 4 units to the right, resulting in the image A' $(5, -2)$. Determine the coordinates of point A .

- 18 In a game of chess, a knight can move around the board in an L -shape. That is, 1 unit in any direction and then 2 units in a perpendicular direction.
If a knight is located at square E now:

- How many possible places could it have been prior to its latest move?
- List all possible places it could have been prior to its latest move.



19 Riccardo has been creating a tower with building blocks.



Is it possible to use the same transformation mapping to place block E or block N at the highest point on the tower? If yes, describe the mapping. If not, explain how you know.

